Module 7 Lecture - Sampling Methods

Research Methods in Psychology

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Table of Contents

1 Week Overview **2**
1.1 Learning Objectives . . . 2
1.2 Week Overview . . . 2

2 Generalizability **2**
2.1 Overview . . . 2
2.2 Relationship Between Populations and Samples . . . 3
2.3 Probability in Sampling Methods . . . 4
2.4 Representative/Probability-based Sampling Types . . . 4
2.4.1 Simple Random . . . 4
2.4.2 Systematic . . . 4
2.4.3 Cluster . . . 5
2.4.4 Multistage . . . 5
2.4.5 Stratified . . . 6
2.4.6 Oversampling . . . 6
2.4.7 Randomness in Sampling and Assignment . . . 7
2.5 Biased Sampling Methods . . . 7
2.5.1 Convenience Sampling . . . 8
2.5.2 Self-sampling/selection . . . 8
2.5.3 Purposeful Sampling . . . 8
2.5.4 Snowball Sampling . . . 9
2.5.5 Quota Sampling . . . 9

3 Checking External Validity **9**
3.1 Overview . . . 9
3.2 Frequency Needs External Validity . . . 10
3.3 When is External Validity Less of a Focus . . . 10
3.4 Larger Sample Does Not Always Equal Better . . . 11

4 Key Points **11**
4.1 Key Points . . . 11

1 Week Overview

1.1 Learning Objectives

- Explain why external validity is essential for most frequency claims.
- Describe which sampling techniques allow generalizing from a sample to a population of interest, and which ones do not.
- Be able to describe the difference between probabilistic and non probabilistic sampling techniques and the resulting representative-ness of the sample
- Understand the risk and implications of biased samples and be able to apply that knowledge to a reading of a real article

1.2 Week Overview

- We can never study _____ in a **population of interest**
  - Example: ***All*** *veterans* or ***all*** *people with GAD*
- Instead, we can only study a smaller _____ of those groups, and then try to extrapolate/generalize our conclusions to the population we care about.
  - Example: Just 40 veterans or 10 people diagnosed with GAD
- That subset that we study on, from the population, is our _____.

📢 Discuss: Think of some other examples of populations and their respective samples

2 Generalizability

2.1 Overview

- A critical notion to consider in how "good" research is in how we sample really relates to the _____ population and variety of settings, i.e.,

external validity

- This consideration is present in pretty much all social research, and in all 3 _____ types (frequency, association, causal)

2.2 Relationship Between Populations and Samples

- To "sample" every individual in a population is called a _____.
  - But such a procedure is not really possible in most research, due to time, money, resources, and ethical _____
  - Example: The US census is (theoretically) a count of _____ people in the United States
  - Instead, we may _____ that an adequately representative sample's behavior is carried over to the population. Thus, a study of a sample's behavior is meant to really be a study of a population's behavior!
- A population of _____ is one that we define when we make a hypothesis. Rarely, do we make a hypothesis or theory applicable to everyone. Instead, we may primarily aim to make a theory _____ around individuals with some specified characteristic(s).
  - Example: In my study, I want to _____ study older adults with suspected dementia (that is my **population of interest**)
  - Example: For that _____ study, I gather 100 older adults from the broader population (that is my **sample**)

📢 Discuss: Think of some of the articles you've read this semester - what were their populations of interest?

- A sample, even if accurately taken from the population of interest, is not inherently _____:
  - **Biased/unrepresentative samples** are those that are somehow taken in a way that they do not properly represent the _____
  - Unbiased/ _____ samples are those that are a proper representation of the population.
  - Realistically, no sample is _____, and we must be discerning in what flaws a sample may have

2.3 Probability in Sampling Methods

- **Probability/random sampling** is all about using a randomization method to _____ members of a population for the sample who then agree or disagree to be in the study
  - Good for _____ validity / generalizability
  - Side note: from a mathematical perspective, the type of random sampling we use is technically _____ due to how computers and chance work, but for the purpose of the class, we will call this random.
- **Non-probability/non-random sampling** is the opposite, in which the method is not _____ and therefore, may be biased towards certain individuals
  - Bad for external validity / _____
- We'll talk about specific types and examples for these in the following sections
- In real studies, we may very well use _____ of the following methods to obtain the final sample to run the study on. There can be a lot a grey area on whether a study used a proper sampling method!

2.4 Representative/Probability-based Sampling Types

2.4.1 Simple Random

- The "gold standard" method, this is if/when we have an equal, known chance of selecting every _____ within a population
- Example: I am interested in a population of GV students, so I put every ID number in a list and then have a computer _____ one at random.
  - Each _____ has a $1/n$ chance to be selected, where n is the number of people in the population
- Fantastic _____, but very difficult and rare

2.4.2 Systematic

- Return to our population of GV students: select two random numbers, say a and b.
  - a will be our a^{th} person in the list of IDs.
  - b will be the number of people we increment by
  - We start with the ath person, and then sample each person b away from that person

- Say we have $a = 4$ and $b = 8$, we would _____ the 4th student, then the 12th, then the 20th, etc on the list

📢 Discuss: Try following the procedure above with a different a and b value (e.g. a = 3, b = 2). What people would you sample?

- Still results in a pretty _____ sample comparable to simple random!

2.4.3 Cluster

- This method comes in when we have naturally existing _____ within our population of interest
  - Example: high school students organized into high schools
- We randomly sample from the clusters (e.g., the high school) and then sample _____ students within a cluster
- Still pretty good!

📢 Discuss: What are some other 'clusters' you can think of that naturally occur?

2.4.4 Multistage

- Simply, cluster random sampling _____ by simple random within the selected clusters. A combination of our descriptions of the prior types.

- Example: I randomly select 10 high schools in the state, and then sample randomly 50 students from a list of students at each school.
- Also, good!

2.4.5 Stratified

- This is a method that is done when we are trying to ensure that our sample is representative of proportions of natural _____ in the population.
- Strata must be some _____ grouping variable (e.g., different ethnicities, different genders, different education levels, etc.) with known levels within the population.
  - E.g., class level (First-year, sophomore, junior, senior) at GVSU
- After we have our stratifying variable, we then randomly sample within each _____ of the variable.
- *However,* our goal is to get a sample with _____ proportions of the strata to the population
  - E.g., If GVSU has 30% First-year, 20% Sophomore, 25% Junior, and 25% senior, I want my sample to have that same breakdown!

2.4.6 Oversampling

- This is a modified stratified sampling, where we _____ represent one or more levels of the strata more in sample, than is found in the population
- This is used when we may want to ensure that we still capture the experience of a relatively _____ group of individuals.
- Example: When gathering my sample I know there will be relatively few indigenous/Native American people. So, I purposefully ensure that at least 10% of the people in my sample are in that group, even though they only represent 5% of the population.

Discuss: What are some other naturally small demographic groups you may encounter in research?

- A related concept for a similar goal is _____ where an unbalanced sample (across some strata) has occurred and we use a statistical technique to give more "value" or weight to less-represented groups.

2.4.7 Randomness in Sampling and Assignment

- Random *sampling* is related to external _____ and how we initially draw our sample from the broader population of people
  - Example: Randomly selecting patients from a hospital for a retroactive study analysis
- Random *assignment* is when we already have our sample and are sorting people into different levels of a manipulated/_____ variable, much more related to experiments and internal validity.
  - Example: after getting my sample, deciding randomly who will get a new trial drug and who will get a placebo during the study

2.5 Biased Sampling Methods

- Biased samples result from poor _____ in the sampling process, which results in the members of the sample being especially "unusual" or those that might behave _____ than other members of the same population
  - Effectively, certain _____ types might mean we are actually gathering individuals of a different population than intended!
- There are _____ issues in doing thorough random sampling techniques
- In some cases, it *may* be _____ to use a method which is *not* random to sample, with some caveats and drawbacks.

Discuss: Take a guess at some 'red flags' of a biased sample. What things would you suspect are telling of bad sampling?

2.5.1 Convenience Sampling

- This occurs when we sample only those people who are _____ to sample
  - Example: Psych 101-mandated research, college students, Amazon MTurk
  - The main problem is that those who are readily _____ may represent individuals who differ from the population of interest - they may be different in motivation, beliefs, status, etc.
- This may also occur _____ in samples that are just hard to track
  - Example: people across a large geographic area
- Be mindful that the method by which we collect data (internet, telephone, etc.) can be un-inclusive and _____ the sample as well.

2.5.2 Self-sampling/selection

- This happens when we rely upon a sample that volunteers or selects itself
  - Example: I put up a poster in the hall with a QR code that says, "take my survey!"
  - Compare this to me sending a randomized email to students of the psychology department
- This is **not** a difference of voluntary vs. involuntary - we still ask for informed consent of everyone regardless of sample type!

2.5.3 Purposeful Sampling

- This is when sampling is done by some method that _____ limits the randomness of who from the population can be sampled, which may be intentional

- Example: Only recruit political science students for a survey by putting posters in the department
  - What about people who don't go into the department or online students.

2.5.4 Snowball Sampling

- When you have current participants _____ other friends / acquaintances for a study.
- This is especially useful when doing research on especially small or b58fc729-690b-4000-b19f-365a4093b2ff-7B7B3C20626C616E6B0A646966666963756C74203E7D7D–to-contact individuals
- Example: Trying to study autistic adults - I may ask my participant to go share my recruitment link with other adults with that diagnosis

2.5.5 Quota Sampling

- Very similar to stratified _____ that we choose individuals non-randomly for each level of our stratifying variable.
- Example: I send out a survey for people to self-sample if they are interested, and I wait until I get at least 15 people of every class level.

Discuss: What are the benefits you see from using these biased methods?

3 Checking External Validity

3.1 Overview

- In my opinion, external validity should _____ be on our mind, because it affects just how much our research means outside the narrow context of the study (external validity)

- *But,* _____ external validity does not necessarily mean that a study is doomed or of no value
- Like any strength or weakness in research, we assess external validity in light of the claim being made. A weaker claim requires _____ robust evidence.
  - Very specific claims with a very _____ population of interest don't need especially high external validity
  - Very broad claims need more evidence of good external validity

Discuss: Do you reckon that most studies do a good job with sampling? Why or why not?

3.2 Frequency Needs External Validity

- Especially when we make a frequency claim about a population, we _____ need robust sampling techniques.
- That is because almost the entire _____ of a frequency claim is on accurate description of the population - and without good external validity, we have nothing.
- Frequency claims are also "more important" when they _____ better across more people. Limits on external validity mean that the claim will not widely apply across too many people.

3.3 When is External Validity Less of a Focus

- As a general rule of thumb, we should *try* to have high external validity wherever possible. Good sampling will always help the meaning and _____ of research.
- *But,* like all validities, we must _____ maximizing generalizability with the practical limits on our resources and time

- In my opinion, the *most* important factor is that researchers transparently report on the _____ of their sampling and are realistic in what this means for the impact of their research.

3.4 Larger Sample Does Not Always Equal Better

- There is a point of _____ returns on sample size to help external validity of a study. Sampling more and more people with a _____ method will still result in a biased sample.
  - Example: after around 100 people, you're not necessarily going to get more representative just by having more people
- Instead, we must focus on both sampling many people, **and** doing this _____ well (i.e., with the probabilistic sampling methods described above)
- Larger samples *will* result in higher statistical _____ (i.e., higher chance of significant findings) - but a significant result does not mean that a study is valid and generalizable!

4 Key Points

4.1 Key Points

- Samples are subsets of the population we are attempting to study. Our conclusions about a sample are meant to represent trends present in the population.
- Sampling methodology is the most important factor in determining external validity and generalizability in a study, aided to a lesser extent by sample size
- Sampling can be done in a randomized manner, resulting in a unbiased sample likely to represent the population. However, it can also be done in an non-randomized way, leading to bias.
- External validity is always important, but tends to be more stressed for frequency claims

The instructor-provided glossary may not include all terms worth memorizing, make sure you consider using the vocabulary list in you book and your own judgment to make sure you have all relevant terms